

ARTIFICIAL INTELLIGENCE BASED CANTEEN FOOD DEMAND PREDICTER USING MACHINE LEARNING

**A mini project report submitted for the partial fulfillment of the requirement for
the award of the degree of**

MASTER OF COMPUTER APPLICATIONS

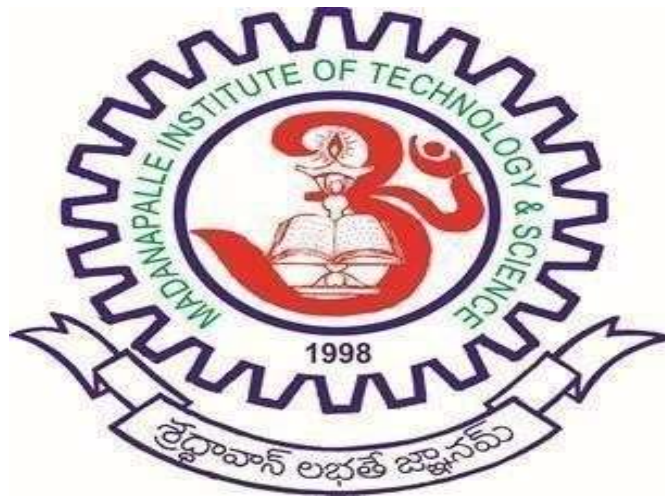
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**DEPARTMENT OF COMPUTER APPLICATIONS
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BONAFIDE CERTIFICATE

This is to certify that the Mini project work entitled “**AI Based canteen food demand predictor using ML**” work carried out by **Hemanth Kumar Patten** (Reg.No.24691F0071), **K. Janardhana** (RegNo.24691F0091), submitted in partial fulfilment of the requirements for the award of the degree of **Master of Computer Applications** in **Madanapalle Institute of Technology and Science, MADANAPALLE**, affiliated to **Jawaharlal Nehru Technological University, Ananthapuramu** during the **academic year 2024-2026**.

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Hemanth Kumar Pattem (Reg.No.24691F0071), K. Janardhana (RegNo.24691F0091), hereby declare that the project entitled “**AI BASED CANTEEN FOOD DEMAND PREDICTER USING ML**” is done by us under the guidance of **Dr. J. Srinivasan Jagannathan, MCA, Ph.D., Assistant Professor**, submitted in partial fulfilment of the requirements for the award of the degree of Master of Computer Applications at **MADANAPALLE INSTITUTE OF TECHNOLOGY & SCIENCE, MADANAPALLE**, affiliated to **Jawaharlal Nehru Technological University Ananthapuramu**, during the academic year 2025-2026. This work has not been submitted by anybody towards the award of any degree.

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ABSTRACT

Food management in college canteens often faces challenges such as excessive food preparation, leading to wastage, or inadequate preparation, resulting in shortages and student dissatisfaction. These issues primarily arise due to the reliance on manual estimation methods, where staff depend on intuition or previous experience to predict daily food requirements. Such approaches fail to consider dynamic factors like fluctuating student attendance, seasonal variations, weekdays, and unexpected events. As a result, canteen operations suffer from inefficiency, financial loss, and poor resource utilization. To address these limitations, the *AI-Based Canteen Food Demand Predictor* is proposed as a modern, intelligent solution that applies Artificial Intelligence (AI) and Machine Learning (ML) techniques to generate accurate food demand forecasts.

The system begins by collecting historical data, which includes daily sales records, student strength, event information, and various temporal factors. This dataset is pre-processed to remove inconsistencies, handle missing values, and normalize patterns for better model learning. Feature engineering is performed to extract the most relevant attributes, such as day-of-week indicators, seasonal patterns, academic schedules, and event-based demand surges. These features play a crucial role in influencing the predictive accuracy of the model.

The core of the proposed system involves training multiple machine learning models to identify patterns in past consumption behaviour. Algorithms such as Linear Regression, Random Forest Regression, and Long Short-Term Memory (LSTM) networks are used due to their capability to handle linear trends, nonlinear relationships, and time-series data respectively. LSTM, in particular, excels at identifying long-term dependencies and recurring patterns in food demand over various intervals. The system evaluates the performance of each model using metrics such as Mean Squared Error (MSE) and Mean Absolute Error (MAE), ensuring that the most reliable prediction approach is selected.

Once trained, the model forecasts the required quantity of food items for upcoming days with significant accuracy. These predictions assist canteen administrators in planning ingredient purchases, regulating preparation quantities, and optimizing daily operations. The system effectively minimizes wastage, reduces unnecessary expenditure, and enhances customer satisfaction by ensuring that food availability aligns closely with demand.

Overall, this project demonstrates how AI-driven analytics can transform traditional canteen management systems into smart, sustainable, and efficient environments. By integrating predictive modelling into routine food preparation processes, institutions can maintain better cost control, reduce environmental impact, and improve operational decision-making. The *AI-Based Canteen Food Demand Predictor* stands as a practical, scalable, and future-ready solution for educational and organizational canteens seeking to modernize their food management strategies.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATED TERM	EXPANSION
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	LSTM	Long Short-Term Memory (networks)
4	MAE	Mean Absolute Error
5	MSE	Mean Squared Error
6	RMSE	Root Mean Squared Error
7	ERD	Entity Relationship Diagram
8	DFD	Data Flow Diagram
9	IoT	Internet of Things

CHAPTER-I

INTRODUCTION

1.1 About The Project

College canteens face a recurring challenge in efficiently managing food preparation. Every day, canteen staff must decide how much of each food item to prepare to meet the needs of students, faculty, and staff. Preparing too much food results in wastage of both resources and money, while preparing too little can lead to dissatisfaction among customers and missed revenue opportunities. This imbalance not only affects the financial performance of the canteen but also impacts overall patron satisfaction. To address this real-world problem, the project titled "AI Based Canteen Food Demand Predictor" has been developed, leveraging Artificial Intelligence (AI) and Machine Learning (ML) techniques to forecast daily food demand accurately.

The primary objective of this project is to provide canteen managers with a reliable system for predicting the quantity of each food item required on any given day. The system achieves this by analyzing historical data, which may include student enrolment numbers, past sales records, weekdays, special events, and seasonal trends. By processing these data points, the system identifies patterns and generates precise predictions, enabling the canteen to prepare the right quantity of food, reduce wastage, and improve operational efficiency.

To ensure accurate predictions, multiple machine learning algorithms are implemented and compared, including Linear Regression, Random Forest, and Long Short-Term Memory (LSTM) networks. Linear Regression helps in understanding simple trends and correlations in sales data, while Random Forest provides robust predictions by considering multiple factors simultaneously. LSTM networks, a type of recurrent neural network, are particularly useful in capturing temporal dependencies and sequential patterns in time-series data such as daily or weekly sales. The performance of the predictive model is evaluated by comparing forecasted demand against actual consumption, ensuring continuous refinement and improvement.

Beyond just demand forecasting, this project contributes to more efficient resource management, cost reduction, and sustainable practices in the canteen. By minimizing food wastage and optimizing preparation, the system not only saves money but also promotes environmentally responsible practices. The project demonstrates a practical application of AI and ML in everyday life, showing how technology can solve common operational challenges in institutions. Overall, the AI Based Canteen Food Demand Predictor represents a creative, intelligent, and impactful approach to improving food management in college canteens.

CHAPTER-II

SYSTEM ANALYSIS

2.1 Existing System

Canteen operations in most educational institutions still follow conventional methods of planning and decision-making, where daily food demand is estimated based primarily on the intuition and experience of the staff rather than being guided by data-driven insights. In this traditional environment, the canteen staff—cooks, supervisors, and managers—tend to rely on past perception instead of systematic analysis or structured forecasting methods. This makes the entire process of determining how much food to prepare imprecise, unreliable, and prone to frequent errors.

Typically, the staff members base their predictions on general observations such as the number of students seen on the previous day or their rough understanding of student turnout patterns. These assumptions become the sole basis for daily planning. However, relying on human judgment alone severely limits the accuracy of these predictions, because human intuition cannot consistently account for the numerous external factors that influence day-to-day food consumption. The demand for food in a canteen fluctuates constantly, depending on academic events, class schedules, weather conditions, student mood, holidays, examination stress, special programmes, and several other parameters. Since most of these variables are not considered in the current system, food preparation lacks strategic planning.

Another major limitation of the existing system is the absence of digital record-keeping. In many canteens, there is no central database that tracks daily sales, the quantity of food consumed, or the usage of raw materials. Without such historical data, it becomes impossible to identify long-term trends, seasonal patterns, or consumption cycles. For example, meal preferences may change during winter or exam seasons, but this information is lost without proper data storage. The lack of structured records prevents administrators from learning from previous mistakes or refining future predictions. Instead, every day's planning becomes isolated, without any meaningful connection to long-term consumption behaviour.

Communication within the canteen workforce is mostly informal and inconsistent. Staff members often decide independently or based on brief verbal instructions from supervisors. If a senior cook is absent or replaced by a new employee, the forecasting ability changes significantly because the predictions are based on personal experience rather than standardized procedures. As a result, the quantity of food prepared may vary drastically from day to day, even when demand remains somewhat stable. This inconsistency affects not only operational efficiency but also the overall service quality experienced by students and staff.

Moreover, the current system does not provide any mechanism for real-time monitoring or dynamic adjustments during the day. Once the food is prepared in the morning, staff cannot adapt to unexpected spikes or drops in student turnout. For instance, sudden rainfall may reduce the number of students visiting the canteen, or an unannounced meeting might bring additional faculty and staff to the cafeteria. Since there is no technological assistance for real-time forecasting or automatic recalculation, the canteen fails to respond to such dynamic situations, resulting in either wastage or shortages.

The existing method also falls short in leveraging modern tools for automation. With no integration of intelligent systems or analytical models, the canteen remains dependent on manual labour for inventory management, demand estimation, and daily planning. Operations such as checking ingredient levels, estimating required quantities, or preparing purchase orders remain slow and inefficient. Without automation, staff members are overburdened, and the entire workflow becomes more susceptible to human errors.

Finally, customer feedback—an important component for improving service—is not incorporated into the planning process. Students may express their dissatisfaction through informal channels, but these opinions

are rarely documented or analyzed. As a result, the canteen is unable to understand dish preferences, meal popularity, or commonly faced service issues. This lack of structured feedback limits improvement opportunities and keeps the system outdated.

In summary, the existing system is constrained by its reliance on subjective judgment, limited consideration of important influencing factors, absence of historical data, lack of real-time adaptability, and minimal use of modern technology. The absence of a data-driven approach leads to frequent mismatches between supply and demand, causing wastage, shortages, and reduced customer satisfaction. A modern canteen requires a more sophisticated, intelligent, and automated framework that can accurately forecast demand, optimize resources, and enhance service quality.

2.1.1 Disadvantages of the Existing System

The limitations of the traditional canteen demand estimation method affect every aspect of the canteen's operation—from food preparation and inventory planning to customer satisfaction and financial outcomes. The major disadvantages are elaborated below, highlighting the critical gaps that necessitate the development of an AI-based predictive system.

1. Reliance on Guesswork and Human Intuition

The most fundamental drawback of the existing system is that it depends entirely on human perception rather than systematic forecasting. Every staff member has a different level of experience, understanding, and judgment. Their predictions can vary widely, leading to inconsistent planning. For example, one cook may assume that student turnout will be high because it is the beginning of the month, while another may feel it will be low due to an upcoming exam. Such subjective assumptions introduce unpredictability and inconsistency, making the entire process unreliable.

Without analytical support, human-based predictions fail to capture the complexities of daily consumption patterns. This lack of accuracy often leads to planning errors that negatively impact the canteen operations.

2. High Probability of Food Wastage

Overestimation of demand is one of the most common consequences of guess-based forecasting. When staff prepare more food than required, large quantities remain unconsumed by the end of the day. This results not only in food wastage but also in unnecessary expenditure on raw materials such as vegetables, grains, spices, and cooking oil.

Furthermore, wastage also leads to wastage of utilities like electricity, gas, and water, which are used extensively in food preparation. The manpower invested in cooking excessive quantities also goes unused. Over time, these losses accumulate and significantly impact the financial sustainability of the canteen.

3. Shortages Due to Underestimation

Underestimation of demand creates equally serious consequences. When food is prepared in insufficient quantities, customers are left without meals, especially during peak hours such as lunchtime. This disrupts service flow and causes inconvenience and dissatisfaction among students and staff.

Shortages damage the reputation of the canteen and reduce customer trust. Students may start avoiding the canteen altogether if they feel that food regularly gets over before they arrive, leading to loss of revenue. Underproduction also creates operational pressure as staff scramble to prepare additional food quickly, often compromising quality.

4. No Consideration of Influential Factors

In the existing system, staff ignore or are unaware of various parameters that significantly affect daily demand. Factors such as:

- Weekday vs. weekend patterns
- Exam schedules
- Weather conditions
- Seasonal meal preferences
- Institutional events or holidays
- Variations in student batches or attendance

All these influence how many people visit the canteen on a given day. Ignoring these factors results in inaccurate predictions. For instance, during exam periods, students often skip lunch or prefer light meals. During winter, demand for hot beverages and snacks increases. Without a mechanism to account for these changes, the existing system remains overly simplistic and highly unreliable.

5. Lack of Historical Data and Trend Analysis

Most canteens do not maintain digital records of daily sales, consumption statistics, or ingredient usage. Without such data, administrators cannot perform basic analyses such as:

- Identifying peak demand days
- Understanding seasonal preferences
- Studying year-on-year changes
- Tracking growth in student population
- Observing reduction or increase in footfall

The absence of trend analysis eliminates the possibility of data-driven decision-making. Staff are unable to refine their forecasting strategy or learn from past outcomes. This keeps the system stuck in a reactive mode where planning is always based on assumptions rather than evidence.

6. Operational Inefficiency

Inaccurate forecasting directly disrupts inventory management. If too much raw material is purchased, it may get spoiled or wasted. If too little is purchased, staff may face shortages while cooking. Similarly, inefficient planning leads to improper workforce allocation, with staff either being underutilized or overburdened.

Poor scheduling, inconsistent preparation times, and mismanagement of resources increase stress among employees and reduce productivity. The lack of automation makes routine tasks time-consuming, and small operational delays accumulate over the day, affecting overall service quality.

7. Financial Losses

Both overproduction and underproduction lead to significant financial losses. When food is wasted, the money spent on raw materials, utilities, and labour is essentially lost. When shortages occur, the canteen misses opportunities to serve customers and generate revenue.

Additionally, poor inventory planning may lead to emergency purchases at higher prices or unnecessary bulk procurement. Over time, these inefficiencies result in increased operational costs and reduced profit margins. For a canteen operating on tight budgets, such losses can be difficult to recover.

8. Inability to Scale or Modernize

The traditional system is rigid and does not adapt well to growth or modernization. If the number of students increases or the menu expands, the current method cannot support the additional complexity. As institutions grow, the canteen must handle larger crowds, more diverse meal options, and faster service expectations.

However, without automation, smart inventory systems, or predictive analytics, scaling becomes difficult. The traditional approach remains suitable only for small-scale operations but collapses under larger or more dynamic environments.

9. Poor Customer Satisfaction

Students expect timely service, sufficient food availability, and consistent quality. When shortages occur or when food is not prepared fresh due to overestimation, the student experience deteriorates. Delays, unavailability of preferred dishes, and lack of variety contribute to negative impressions.

Over time, poor customer satisfaction affects the canteen's reputation. Students may prefer outside food options or avoid visiting the canteen altogether, leading to a loss in footfall and revenue.

10. No Feedback Integration

The current system does not include any structured mechanism for collecting, storing, or analyzing customer feedback. Student opinions about taste, quantity, service timing, or menu preferences are not formally documented. Without capturing real user behavior, the canteen cannot adapt to evolving needs or preferences.

Feedback is essential for understanding daily demand, identifying popular dishes, and improving operational decisions. The absence of this database keeps the planning process outdated and disconnected from students' actual needs.

2.2 Proposed System

The proposed AI-Based Canteen Food Demand Prediction System is developed as an advanced, intelligent, automated, and highly reliable alternative to the existing manual forecasting approach used in most canteens. Where the traditional system relies heavily on guesswork and subjective human judgment, the proposed system introduces a structured framework based on machine learning, predictive modeling, data analytics, and systematic data management. It transforms unorganized raw information into actionable insights that enable administrators to prepare accurate quantities of food, optimize operational efficiency, and minimize both wastage and shortages.

At its core, the proposed system aims to revolutionize canteen planning through the integration of modern data science methods. The system begins by collecting diverse historical datasets that affect food consumption patterns. These include daily food sales records, student and staff population data, day-of-week information, seasonal influences, academic schedules, holiday calendars, special event schedules, weather conditions, and monthly variations in attendance. By incorporating such a wide variety of features, the system moves far beyond the simplicity of the traditional method. Instead of relying on a single variable such as previous day's turnout, it considers a multidimensional set of factors that collectively shape food demand.

2.2.1 Data Collection and Dataset Structuring

Data collection acts as the foundational layer of the proposed system. Without comprehensive datasets, machine learning models cannot learn meaningful patterns. The system gathers information from multiple sources including:

- **Daily consumption logs** (number of plates, items sold, peak-hour usage)
- **Student and staff count data** (present strength, attendance variations)
- **Day and date indicators** (weekday/weekend, start/end of month)
- **Academic events** (examinations, seminars, workshops, sports events)
- **Holiday and festival calendars**
- **Weather conditions** (rainy days typically reduce outdoor movement)
- **Monthly and seasonal variations** (summer/winter eating trends)

These datasets are stored in a structured database, ensuring consistency and accessibility for training machine learning models. The more historical data collected, the better the model becomes at generating precise forecasts.

2.2.2 Data Preprocessing and Cleansing

Raw data often contains noise, missing values, and inconsistencies. Therefore, the system performs a rigorous data preprocessing phase to ensure the reliability of the learning process. This stage includes:

✓ Handling Missing Values

Missing entries are replaced using techniques such as mean/median imputation, interpolation, or logical approximation based on the context.

✓ Removing Duplicate Records

Repeated entries are identified and removed to prevent model bias.

✓ Encoding Categorical Variables

Since machine learning models require numerical input, descriptive values such as day names, weather types, or holiday indicators are converted into numeric formats using techniques such as one-hot encoding or label encoding.

✓ Normalization/Standardization

Features with large numeric ranges (e.g., attendance, temperature) are scaled to uniform ranges to improve model performance.

✓ Outlier Detection

Sudden spikes or dips in consumption due to one-time events are carefully analyzed and treated appropriately.

The result of this preprocessing stage is a clean, well-structured dataset capable of supporting advanced analytics and predictive modeling.

2.2.3 Feature Selection and Feature Engineering

Identifying the most relevant features is essential for enhancing prediction accuracy. The system evaluates multiple variables to determine which ones significantly influence food consumption. Some key features selected include:

- Weekday vs. Weekend trends
- Student and staff attendance
- Academic and cultural events
- Examination schedules
- Weather patterns
- Holiday proximity
- Seasonal preferences
- Special menu items
- Festive periods

Feature engineering further enhances the dataset by generating new derived attributes—for example:

- "Is_Exam_Week" flag
- "Is_Rainy_Day" indicator
- Rolling averages of past 7-day or 30-day consumption
- Lag features (previous day/previous week demand)
- Month-wise grouping variables

These engineered features help the model learn deeper, more meaningful relationships within the data.

2.2.4 Machine Learning Model Training

Once the dataset is prepared, the system moves to the model training stage. Several machine learning and deep learning models are employed to determine which approach delivers the most accurate and reliable predictions. Some of the key models used include:

◆ Linear Regression

A simple model used as a baseline to understand linear trends.

◆ Random Forest Regressor

An ensemble of multiple decision trees, effective in handling diverse features and reducing overfitting.

◆ LSTM (Long Short-Term Memory Networks)

A powerful deep learning model designed for time-series forecasting that learns long-term dependencies and patterns in sequential data.

Each model is trained using the historical dataset and evaluated using metrics such as:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

The dataset is split into training and testing subsets to eliminate evaluation bias and ensure real-world reliability. Cross-validation techniques are applied to further validate the robustness of the models.

After comparing model performance, the algorithm with the highest accuracy and lowest error values is selected as the final prediction model. In many cases, LSTM or Random Forest models provide superior results due to their ability to capture complex relationships.

2.2.5 Prediction Engine and Output Generation

Once trained, the selected model becomes the main prediction engine of the system. It can generate:

- **Daily predictions** (e.g., expected consumption for tomorrow)
- **Weekly predictions** (useful for scheduling and planning)
- **Item-specific predictions** (rice, chapati, curry, snacks, etc.)
- **Event-based predictions** (higher demand during festivals or sports events)

These predictions are generated automatically, and the system presents them through a user-friendly interface.

2.2.6 User Interface and Dashboard Design

The proposed system includes a visually rich, intuitive dashboard that enables easy interpretation of predictions. Key features include:

- **Charts and Graphs:** Trend lines, bar charts, heatmaps
- **Daily Food Prediction Panels:** Display exact quantities required
- **Interactive Filters:** Users can select specific dates, events, or food items
- **Reports and Data Export:** Predictions can be downloaded as PDF/Excel
- **Admin Settings:** Allows management of users, database, and model updates

The dashboard serves as the central console through which canteen staff and administrators can monitor food demand quickly and effectively.

2.2.7 Real-Time Data Integration and Adaptive Learning

The system supports dynamic updating of data. As new daily records are added, the machine learning model continuously learns and adapts to emerging consumption patterns. This ensures:

- Higher accuracy over time
- Ability to capture new seasonal or behavioural trends
- Improved sensitivity to sudden demand changes

Real-time integration helps make the model responsive and context-aware.

2.2.8 Backend Integration and System Architecture

The proposed system adopts a modular architecture consisting of:

1. Data Collection Module
2. Data Preprocessing Module
3. Feature Engineering Module
4. Model Training and Selection Module
5. Prediction Engine
6. Visualization and Dashboard Interface

Such a design makes the system flexible, scalable, and easy to modify. If new technologies such as IoT sensors or cloud databases are introduced in the future, they can be integrated seamlessly without redesigning the entire system.

2.2.9 Practical Outcomes and Operational Improvements

The overall goal of the proposed system is to modernize canteen operations through intelligent automation. With accurate forecasts, the canteen can:

- Reduce unnecessary wastage
- Avoid shortages during peak hours
- Improve inventory management
- Optimize staff scheduling
- Reduce operational costs
- Enhance customer satisfaction

Thus, the proposed system becomes not only a technological solution but also a vital operational tool.

2.2.1 Advantages of the Proposed System

The adoption of this AI-based prediction model brings several direct and indirect advantages. Each benefit contributes to improving operational efficiency, reducing costs, and enhancing customer satisfaction.

1. Highly Accurate, Data-Driven Predictions

The proposed system uses machine learning models that analyze historical consumption patterns, variable dependencies, and multi-factor influences to deliver highly accurate forecasts. Unlike human intuition, the system remains unbiased and analytical, ensuring consistent and reliable predictions. The accuracy increases over time as more data is accumulated.

2. Major Reduction in Food Wastage

By forecasting exactly how much food is required on a given day, the system helps avoid overproduction. Every meal prepared aligns closely with predicted demand, which significantly reduces leftovers, spoilt food, and unnecessary usage of raw materials. This not only benefits the environment by minimizing food wastage but also reduces operational costs.

3. Prevention of Shortages

Underproduction is one of the biggest issues in manual estimation. The proposed system minimizes shortages by forecasting real demand accurately, ensuring sufficient food is prepared even on unpredictable busy days. This leads to smoother service, timely meal delivery, and happier customers.

4. Automated and Error-Free Processing

The system automates tasks such as data cleaning, feature extraction, model training, and prediction generation. Automation reduces human involvement in repetitive processes, minimizing errors and speeding up the workflow. This ensures that decisions are based on precise and reliable values rather than subjective judgment.

5. Improved Administrative Decision-Making

The prediction dashboard provides administrators with detailed charts, history logs, trend analyses, and consumption reports. These insights help with:

- Menu planning
- Inventory purchase decisions
- Budget allocation
- Workforce management
- Monitoring student behaviour trends

Data-driven decision-making enhances the overall strategic functioning of the canteen.

6. Adaptability to Changing Conditions

The system continuously updates its knowledge base as new data is added. It adjusts its predictions based on:

- Seasonal preference shifts
- Academic schedule changes
- Sudden rise/fall in attendance
- Unexpected events

This dynamic adaptability makes the system robust and relevant throughout the year.

7. Efficient Utilization of Resources

With accurate predictions, the canteen can:

- Purchase only necessary raw materials
- Allocate staff workload correctly
- Avoid emergency purchases
- Reduce wear and tear on equipment
- Utilize manpower appropriately

This leads to smoother operations, reduced cost, and efficient time management.

8. Scalable and Easy to Upgrade

The architecture of the system is modular, meaning new features can be added without impacting existing modules. Future upgrades may include:

- IoT-based automated attendance reading
- Mobile apps for canteen staff
- Real-time cloud analytics
- Integration with payment or billing systems

This scalability ensures long-term usability and technological relevance.

9. Enhanced User Convenience

The system offers a clean and user-friendly interface. Even staff with minimal technical skills can easily access predictions, view trends, and interpret results. The simplicity of the dashboard, coupled with the richness of insights, enhances daily operations.

10. Financial Savings and Sustainability

By minimizing wastage, optimizing resource use, and improving planning, the system significantly reduces operational expenses. These savings can contribute to better pricing, improved quality, or investment in infrastructure. Additionally, reduced wastage supports sustainability goals and responsible resource utilization.

CHAPTER-III

HARDWARE AND SOFTWARE REQUIREMENTS

The successful implementation of the **AI-Based Canteen Food Demand Prediction System** requires a combination of efficient hardware components and reliable software tools. Since the project involves data preprocessing, machine learning model training, and visualization, the hardware and software must be capable of handling computational workloads smoothly. The following specifications outline the minimum and recommended requirements for developing, training, and deploying the prediction system.

3.1 Hardware Requirements

1. **Processor: Intel Core i3/i5 or Higher**

A multi-core processor is essential for executing machine learning algorithms, preprocessing large datasets, and running the application without lag. Intel i5 or higher provides faster computation, especially during model training and testing phases.

2. **RAM: Minimum 4 GB (8 GB Recommended)**

Memory influences the speed of data processing and training. While 4 GB RAM is sufficient for basic model development, 8 GB or more is recommended for smoother execution, especially when using deep learning models like LSTM.

3. **Storage: 250 GB HDD / SSD**

Adequate storage is required to maintain datasets, logs, trained model files, and development tools. Using an SSD significantly improves read/write speeds, reducing data loading time and enhancing overall performance.

4. **System Type: 64-Bit Operating System**

A 64-bit architecture ensures compatibility with machine learning libraries and supports efficient memory handling during training and data processing tasks.

5. **Graphics: Integrated Graphics (GPU Optional)**

Basic integrated graphics are enough for traditional ML models. However, an NVIDIA GPU (e.g., GTX Series) can speed up training for deep learning models like LSTM, though it is not mandatory.

3.2 Software Requirements

1. **Operating System: Windows 10/11 or Linux**

The project can be developed on both Windows and Linux operating systems. Linux distributions like Ubuntu are preferred for machine learning due to better environment stability and compatibility with ML frameworks.

2. **Programming Language: Python 3.x**

Python is chosen because of its vast machine learning libraries, ease of use, and community support. Python 3.x ensures compatibility with modern ML frameworks.

3. **Libraries and Frameworks:**

- **NumPy** – For efficient numerical computations, matrix operations, and data manipulation.
- **Pandas** – Used for dataset handling, cleaning, preprocessing, and organizing structured data.
- **Matplotlib / Seaborn** – Essential for generating graphs, visualizing trends, and creating insights from food consumption data.
- **Scikit-learn** – Provides machine learning algorithms like Linear Regression, Random Forest, and evaluation metrics.
- **TensorFlow / Keras** – Required for designing, training, and deploying deep learning models such as LSTM for time-series prediction.

4. **IDE / Development Tools: Jupyter Notebook / PyCharm / VS Code**

These tools offer a user-friendly environment for coding, debugging, model visualization, and running experiments. Jupyter Notebook is particularly useful for step-by-step ML development.

5. **Database (Optional): MySQL or SQLite**

Used to store historical consumption records, daily entries, prediction logs, and model outputs. SQLite is suitable for small-scale systems, while MySQL supports larger deployments.

6. **Version Control (Optional): GitHub / Git**

Version control tools help track project development, manage code changes, and enable team collaboration. GitHub also provides cloud backup and issue tracking.

CHAPTER-IV

SYSTEM DESIGN

4.1 Module Description

The AI-Based Canteen Food Demand Prediction System is designed using a modular architecture to ensure scalability, ease of maintenance, and efficient integration of machine learning components. Each module performs a specific function and interacts seamlessly with others to deliver accurate predictions and user-friendly insights. The following sections describe each module in-depth.

1. User Module (Admin/Canteen Staff Login)

This module is responsible for managing secure access and ensuring that only authorized personnel can interact with the system. It includes a login interface where users enter their username and password. The module verifies credentials against stored records using authentication techniques such as hashing and role verification. The system supports multiple roles:

- **Admin** users have full privileges, including data upload, model training, viewing analytics, managing users, and configuring system settings.
- **Canteen Staff** users have limited access, primarily focused on entering daily sales data and viewing predicted quantities.

The module also manages user session handling, password encryption, and logout functionality. Furthermore, it includes profile management features where users can update basic details, change passwords, and track system usage logs. This module ensures a secure and controlled environment for operating the prediction system.

2. Data Collection Module

The Data Collection module acts as the backbone of the prediction system. It gathers all relevant data required to train machine learning models and generate accurate forecasts. The data sources may include:

- **Daily sales entries** for each food item
- **Student and staff attendance records**
- **Day-wise data** such as weekdays, weekends, and holidays
- **Special events and functions** that affect food consumption
- **Weather or temperature data** (optional but useful)

This module ensures that data is consistently recorded and stored in a well-structured database. It may support manual data entry through the user interface or automated data import from external files like CSV or Excel. By standardizing data formats and maintaining accurate historical logs, the module ensures that downstream preprocessing and prediction tasks receive high-quality, reliable input.

3. Data Preprocessing Module

Raw data often contains inconsistencies that can negatively affect the performance of machine learning models. The Data Preprocessing module resolves these issues by performing systematic cleaning and transformation tasks. Its operations include:

- **Handling missing values** through techniques like mean/median replacement or interpolation
- **Removing duplicate entries** to avoid bias
- **Encoding categorical data** such as “Weekday” or “Event” into numerical values
- **Normalizing or scaling numerical features** (e.g., using MinMaxScaler for LSTM)
- **Detecting outliers** and managing unusual spikes in data
- **Organizing data sequences** for time-series models

By converting the raw, inconsistent dataset into a clean and standardized format, this module significantly improves model efficiency and prediction accuracy.

4. Machine Learning Model Training Module

This module focuses on developing predictive models using historical data. It integrates multiple machine learning algorithms to ensure comprehensive comparison and optimal accuracy. The key steps include:

- **Splitting the dataset** into training and testing sets
- **Training models** like Linear Regression, Random Forest, and LSTM
- **Hyperparameter tuning** to improve the model’s learning capability
- **Evaluating performance** using metrics such as MAE, MSE, RMSE, and R^2 score
- **Selecting the best model** based on accuracy and error values
- **Saving the trained model** so it can be reused without retraining

For deep learning models like LSTM, the module prepares sequential data inputs (e.g., last 7 days of features) and trains the architecture to capture long-term dependencies. This module forms the core intelligence of the system and continuously improves as more data becomes available.

5. Prediction Module

Once a model is trained, the Prediction Module uses it to estimate the quantity of food required for upcoming days. This module takes various input features such as:

- Student strength
- Weekday or weekend indicator
- Holiday or event data
- Temperature or weather information
- Any additional dynamic factors

It processes the input through the trained model and generates predicted demand values for each food item. These predictions help canteen staff plan meals accurately, avoid overproduction, reduce wastage, and ensure sufficient availability. The module can also run in automatic scheduling mode, where it generates predictions daily without manual input.

6. Report Generation Module

This module provides comprehensive analytical insights to support decision-making. It automatically generates various types of reports, including:

- **Daily prediction reports** showing estimated vs. actual consumption
- **Trend analysis charts** for weekly, monthly, or seasonal demand patterns
- **Accuracy graphs** indicating model performance
- **Food wastage reports** highlighting surplus or shortages
- **Comparative analysis** between different models
- **Inventory planning suggestions** based on predicted values

Reports can be displayed as tables, line charts, bar graphs, or downloadable documents. This module is essential for administrators who rely on data-driven insights to optimize canteen operations, reduce costs, and enhance service quality.

4.2 Tables

Date	Day	Weather	Category	Food_Item	Temperature	IsWeekend	IsHoliday	Quantity_Sold
16-07-2025	Wednesday	Rainy	Veg	Veg Biryani	27.7	0	0	88
08-09-2025	Monday	Cloudy	Non-Veg	Egg Rice	28.3	0	0	134
13-09-2025	Saturday	Cloudy	Non-Veg	Chicken Noodles	27.8	1	0	148
19-07-2025	Saturday	Sunny	Non-Veg	Egg Curry	23.6	1	0	30
24-08-2025	Sunday	Sunny	Non-Veg	Chicken Biryani	29.7	1	1	21
15-07-2025	Tuesday	Cloudy	Non-Veg	Egg Curry	24.4	0	0	122
23-07-2025	Wednesday	Cloudy	Veg	Paneer Curry	24.2	0	0	39
07-07-2025	Monday	Rainy	Veg	Dokla	29	0	1	143
26-09-2025	Friday	Sunny	Veg	Dokla	31.8	0	0	37
01-07-2025	Tuesday	Cloudy	Non-Veg	Egg Curry	29.9	0	0	137
07-08-2025	Thursday	Cloudy	Veg	Dokla	26.6	0	0	93
21-07-2025	Monday	Sunny	Veg	Mushroom Curry	29.9	0	0	52
24-08-2025	Sunday	Cloudy	Non-Veg	Egg Rice	30.9	1	1	33

Fig 1. Dataset

4.3 Use Case Diagram

Actors:

- **Admin** – Responsible for managing data, configuring the system, and viewing reports.
- **Staff** – Enters daily food sales, student count, and views daily predictions.
- **ML Engine** – Processes data, builds models, and generates predictions.

Use Cases:

- Login
- Manage Food Menu
- Enter Sales Data
- Upload Student Strength
- Train ML Model
- View Predictions
- Generate Reports

These use cases represent the interaction between users and the system for data management, prediction generation, and reporting.

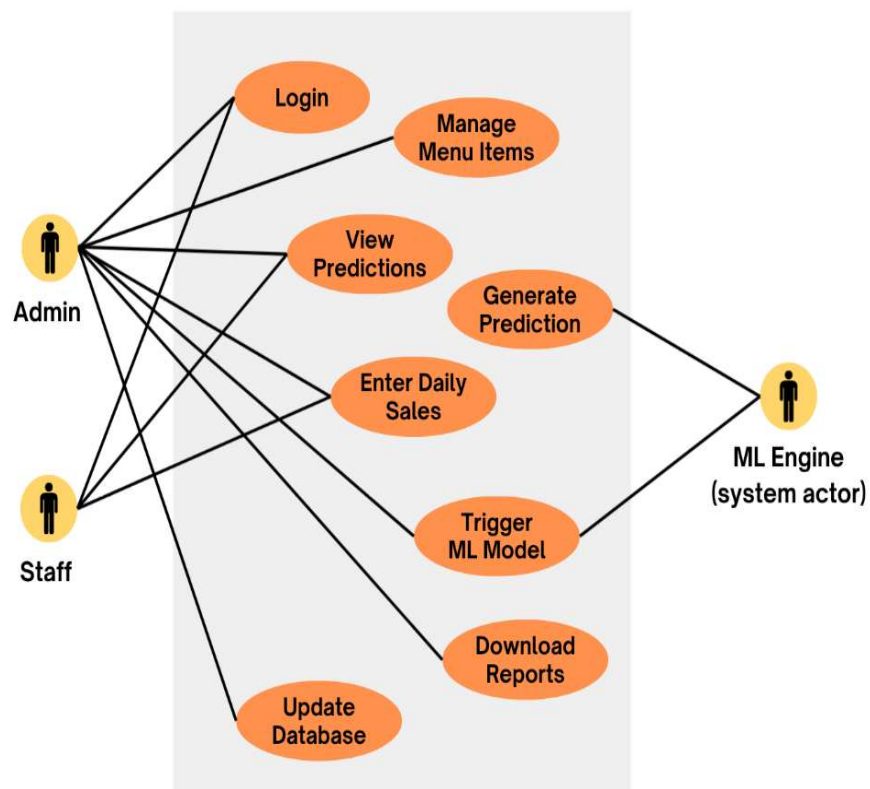


Fig 2. Use Case Diagram

4.4 Class Diagram

Class Name	Attributes	Methods
User	userID, username, password, role	login(), manageData()
FoodItem	itemID, itemName, category, cost	addItem(), updateItem()
SalesData	salesID, date, itemID, students, qtySold, event	recordSales(), fetchSales()
MLModel	modelID, modelType, accuracy	trainModel(), testModel(), saveModel()
Prediction	predictionID, date, itemID, predictedQty	generatePrediction(), compareActual()

This diagram shows the relationship between system classes like User, FoodItem, SalesData, MLModel, and Prediction. It visualizes system data structure and method interactions.

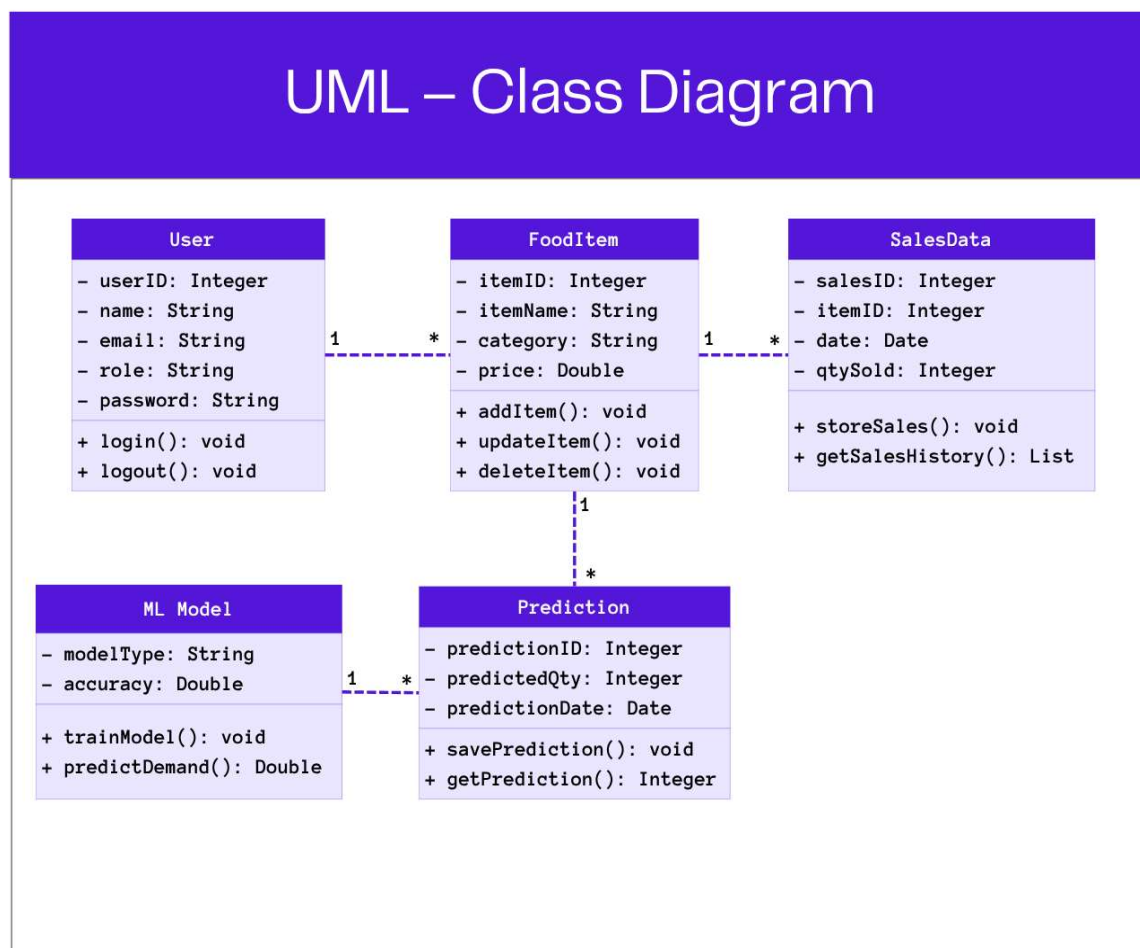


Fig 3. Class Diagram

4.5 ER Diagram

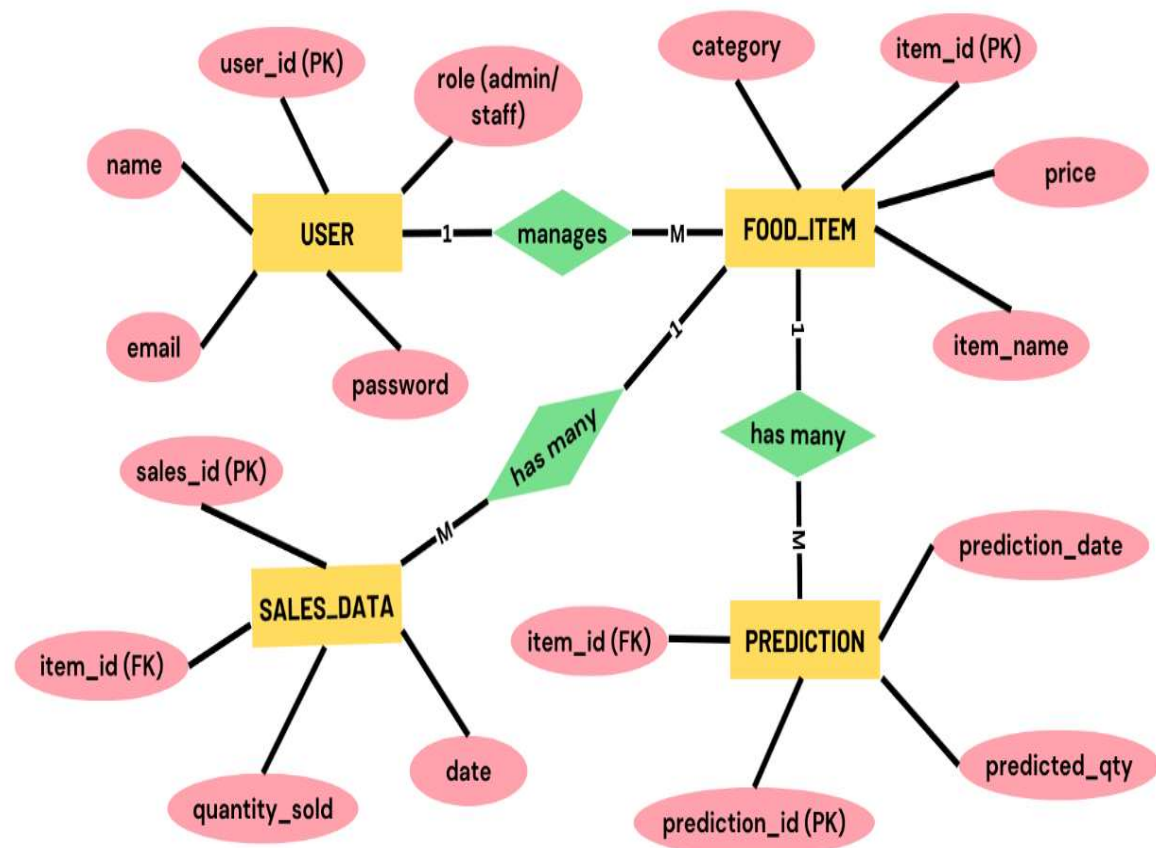


Fig 4. ER Diagram

1. User Entity

- **Primary Key:** userID
- **Attributes:** username, password, role
- **Purpose:** Stores login details and access roles of admin/staff who use the system.
- **Role in System:** Responsible for entering sales data, viewing predictions, and managing food records.

2. Food_Item Entity

- **Primary Key:** itemID
- **Attributes:** item Name, category, price
- **Purpose:** Contains information about food items available in the canteen.
- **Usage:** Helps in tracking item-wise sales data and predictions.

3. Sales_Data Entity

- **Primary Key:** saleID
- **Foreign Key:** itemID (from Food_Item)
- **Attributes:** date, students, qtySold, event
- **Purpose:** Stores historical sales records, including how many students attended and the quantity of each item sold per day.
- **Usage:** Used as training data for machine learning models.

4. Prediction Entity

- **Primary Key:** predictionID
- **Foreign Key:** itemID (from Food_Item)
- **Attributes:** date, predictedQty
- **Purpose:** Stores the predicted demand (quantity) for different food items on upcoming days.
- **Usage:** Helps canteen staff prepare accurate food quantities.

Relationships in ER Diagram

Relationship Type	Description
1 User → Many Sales_Data	One user (Admin/Staff) can enter multiple sales records.
1 Food_Item → Many Sales_Data	One food item may appear in many sales entries.
1 Food_Item → Many Prediction	One food item may have multiple future demand predictions.

4.6 Data Flow Diagram (DFD)

DFD – Level 0 (Context Level)

User/Canteen Staff



AI-Based Canteen Food Demand Prediction System



Predicted Food Quantity / Report Output

Explanation:

- The user inputs data such as sales records, student count, and events.
- The system processes this information using AI models.
- It generates predicted food demand for the upcoming day.

DFD – Level 1

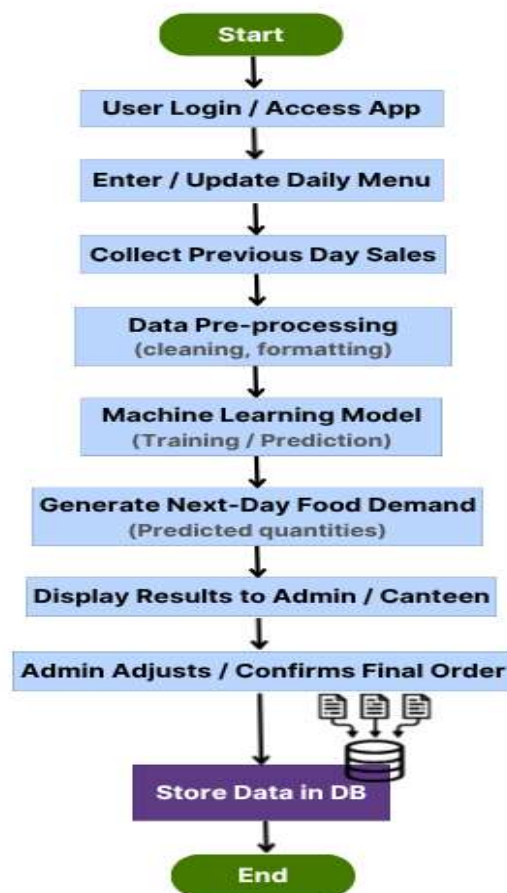


Fig 5. Food Demand Prediction System Workflow

4.7 Technology Stack

Programming & Development

Component	Technology Used
Programming Language	Python 3.x
Development Tools	Jupyter Notebook, PyCharm, VS Code
Version Control (Optional)	Git / GitHub

Machine Learning Libraries

Purpose	Libraries
Numerical Processing	NumPy
Data Preprocessing	Pandas
Visualization	Matplotlib, Seaborn
Machine Learning Models	Scikit-learn (Random Forest, Linear Regression)
Deep Learning	TensorFlow / Keras (LSTM)

Database & Storage

Type	Technology
Database	MySQL / SQLite
Dataset Format	CSV, Excel, SQL Tables

System Requirements

Requirement	Specification
RAM	Minimum 4 GB (8 GB recommended)
Processor	Intel i3/i5 or higher
OS	Windows 10/11, Ubuntu Linux
Storage	250 GB HDD / SSD

CHAPTER-V

SYSTEM IMPLEMENTATION

System implementation is the phase where the conceptual design converts into a fully functional, operational system. In the AI-Based Canteen Food Demand Prediction System, implementation involves integrating hardware, software, machine learning models, user interface, database, and backend services to create a cohesive, reliable environment. This section explains how each component of the system is implemented, how the modules interact, and how the selected technologies support smooth functioning.

Implementation begins with setting up the development environment, installing required libraries, configuring the database, and preparing datasets. This preparation ensures that the system runs efficiently, supports data processing, and allows machine learning models to train effectively. The entire system follows a modular approach, where each module—such as User Management, Data Preprocessing, Model Training, Prediction Generation, and Report Display—is implemented independently to enhance maintainability and scalability.

The User Interface (UI) is implemented using simple and intuitive web technologies. HTML and CSS provide a clean layout, while JavaScript enhances user interactivity. The UI allows staff and administrators to log in, enter daily food consumption data, view predictions, and generate reports. The system also includes form validations to prevent incorrect or incomplete data entry. These features ensure that even users with minimal technical knowledge can use the system easily.

The backend layer forms the core of the system implementation. It handles major operations such as data storage, data transformation, model execution, and communication between the UI and the machine learning engine. Python is used to implement backend logic due to its strong support for data analysis and machine learning. The backend interacts with the database to fetch historical records, update new entries, store prediction outcomes, and retrieve data for visualization. A well-structured backend ensures fast data flow and seamless user experience.

The machine learning implementation is the most crucial part of the system. Historical sales data undergoes preprocessing steps such as handling missing values, normalization, feature encoding, and splitting into training-testing sets. Multiple machine learning algorithms are implemented, such as Linear Regression, Random Forest, and LSTM networks. The models are trained to learn patterns in daily consumption, seasonal trends, student strength, weekdays, and event impacts. After training, the model with the highest accuracy and lowest error metrics is selected as the prediction engine. The trained model is saved using serialization techniques such as Pickle or Joblib so that it can be loaded and used instantly without retraining each time.

To perform predictions, the backend loads the trained model and passes input features (such as date, weekday, attendance, and previous sales). The model outputs the predicted quantity for each food item, and these results are displayed on the UI through charts and tables. For visualization, libraries like Matplotlib and Seaborn are used, providing clear graphs that help administrators easily interpret the results. The report generation module compiles prediction summaries, daily trends, and historical comparisons into structured reports.

Finally, the system undergoes integration, where all modules (UI, backend, database, ML model, reports) are combined into a single functional platform. The system is tested for correctness, performance, usability, and load capacity. Once validated, the system becomes ready for deployment in the canteen environment, where staff can use it daily to forecast food requirements accurately.

5.1 Language Selection

The choice of programming languages and technologies plays a critical role in ensuring efficiency, flexibility, reliability, and ease of implementation. In this project, Python is selected as the primary development language due to its unmatched support for machine learning and data processing. Along with Python, technologies such as HTML, CSS, JavaScript, MySQL, and various ML libraries are used to build the frontend, backend, and database layers. The selection of each language is based on performance, compatibility, ease of use, and suitability for the specific tasks involved in the system.

1. Python as the Core Programming Language

Python is the backbone of system implementation and machine learning. It is used for:

- Data preprocessing
- Machine learning model training
- Prediction computation
- Visualization
- Backend logic
- API integration

Python is chosen for several reasons:

a. Strong Machine Learning Support

Python offers a rich ecosystem of ML libraries such as Scikit-learn, TensorFlow, Keras, NumPy, and Pandas. These libraries simplify the process of building models, evaluating performance, and making predictions.

b. Easy to Learn and Read

Python has a clean, readable syntax that makes development faster and reduces code complexity. This is beneficial for both beginners and professionals.

c. Large Community and Documentation

Python has one of the largest developer communities in the world. This ensures extensive documentation, quick bug fixes, and availability of resources for problem-solving.

d. Rapid Prototyping

Python allows developers to quickly test ideas, build prototypes, and refine models without excessive coding workload.

e. Cross-platform Compatibility

Python works smoothly on Windows, macOS, and Linux, making it ideal for diverse environments.

Overall, Python is the perfect choice for developing a prediction-based system where heavy computation and data analysis are involved.

2. HTML and CSS for Interface Design

HTML and CSS are used for designing the user interface. They provide a clear structure, visual formatting, and user-friendly layout.

a. HTML (HyperText Markup Language)

HTML creates the core structure of web pages such as login page, dashboard, data entry forms, tables, and report views. It is simple to use and supports interaction with backend scripts.

b. CSS (Cascading Style Sheets)

CSS defines the visual appearance — colors, spacing, buttons, menu design, fonts, and alignment. Using CSS ensures consistency in design and professional appearance.

These languages ensure the system looks clean, modern, and easy to navigate.

3. JavaScript for Dynamic Interaction

JavaScript is used to add interactivity and enhance user experience. It supports:

- Form validation
- Dynamic updates
- Interactive charts
- UI responsiveness

JavaScript helps the system become more intuitive by enabling real-time interactions without reloading pages.

4. MySQL or SQLite for Database Management

A database is necessary to store historical consumption records, predictions, user information, and logs. MySQL and SQLite are popular due to:

a. Excellent performance for structured data

They efficiently store and query large datasets used for machine learning.

b. Reliability and data integrity

Transactions ensure data consistency.

c. Easy integration with Python

Libraries like MySQL-Connector and SQLite3 enable seamless connectivity.

d. Scalability

MySQL supports large-scale deployments, whereas SQLite works well for standalone systems.

5. Machine Learning Libraries

The ML libraries selected significantly influence model accuracy and system performance.

a. NumPy

Supports fast numerical computation using arrays and matrices.

b. Pandas

Used for data cleaning, feature extraction, merging, and transformation.

c. Scikit-learn

Offers ML algorithms like Linear Regression, Random Forest, Decision Trees, and evaluation tools such as MAE, RMSE, and R^2 .

d. TensorFlow / Keras

Used for implementing advanced deep learning models such as LSTM, which handle time-series forecasting efficiently.

e. Matplotlib / Seaborn

Used for visualizing trends, patterns, and final predictions in chart form.

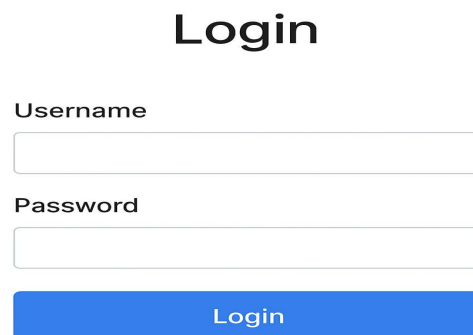
6. IDEs and Development Tools

The system is implemented using tools such as Jupyter Notebook, PyCharm, and VS Code. These tools provide:

- Code suggestions
- Built-in debugging
- Interactive development
- File management
- Easy plugin integration

Jupyter Notebook is particularly useful for building ML models step-by-step and visualizing output instantly.

5.2 Screenshots



The screenshot shows a login interface with the title 'Login' centered at the top. Below the title, there are two input fields: the first is labeled 'Username' and the second is labeled 'Password'. Both fields are empty and have a light gray border. Below the password field is a blue button with the text 'Login' in white, centered on the button.

Fig 6. User login page

Dashboard

Dashboard

Data Entry

Prediction

Reports

Dashboard

Today's Sales

Food Item	Prediction Output
Fried Rice	160
Noodles	120
Veg Curry	130
Paneer Curry	100
Dal	85

Prediction Output

Fig 7. Dashboard

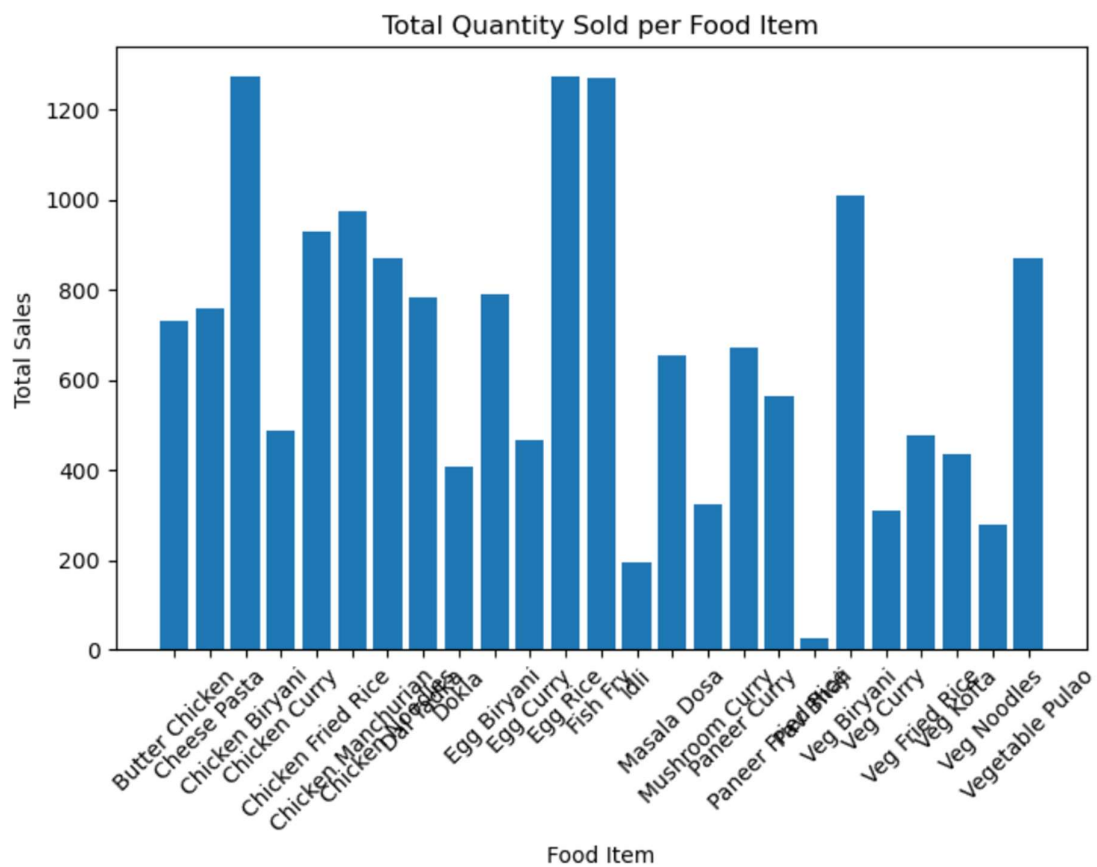


Fig 8. Total Quantity Sold per Food Item

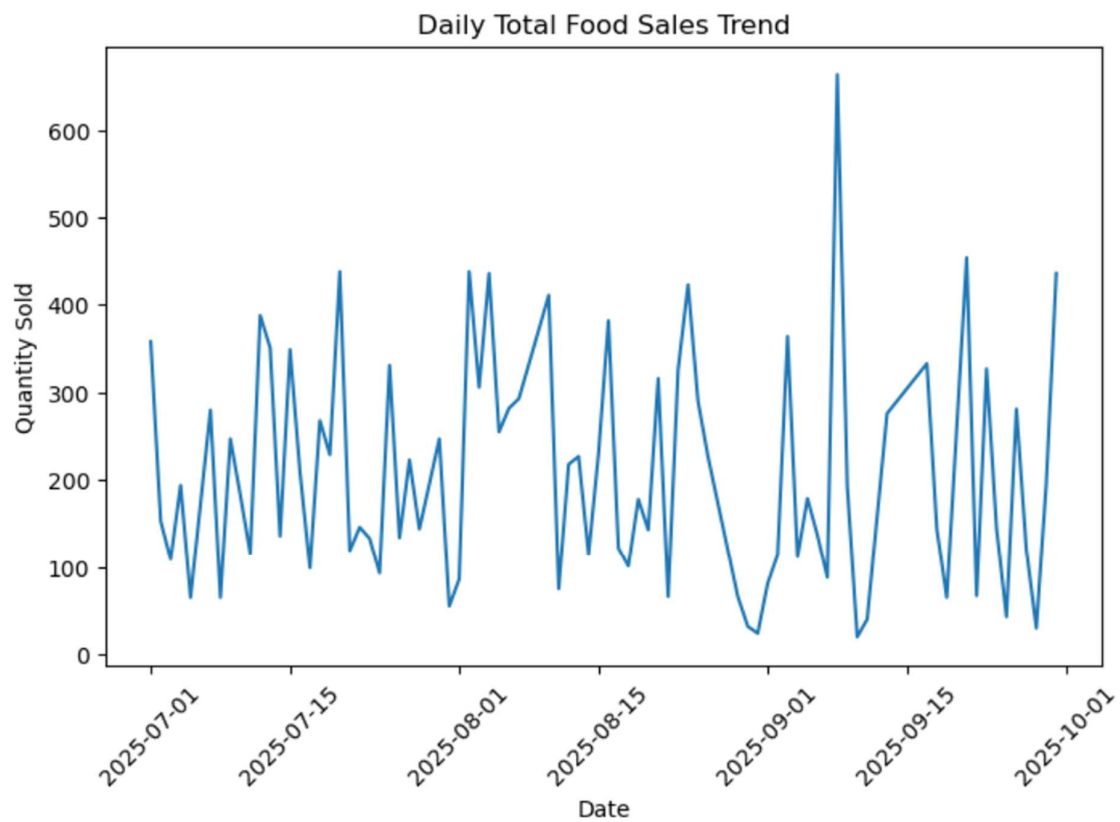


Fig 9. Daily Food Sales Trend

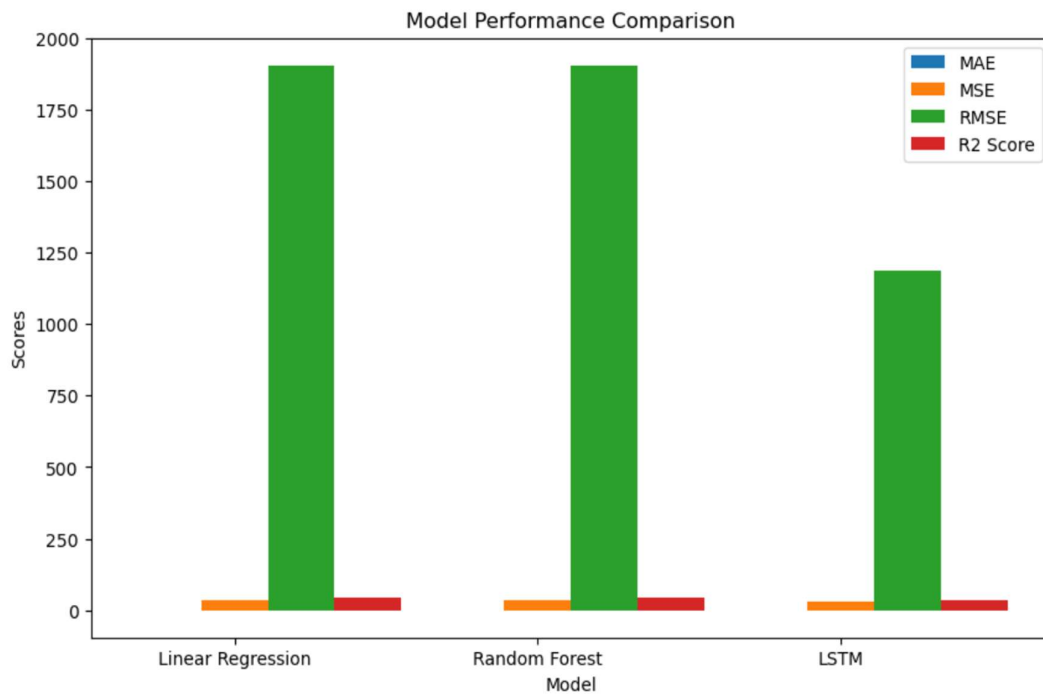


Fig 10. Machine Learning Model Performance Comparison

5.3 Sample Code

4.3.1 Models Used and Feature Description

In the development of the AI-based Canteen Food Demand Predictor, a combination of Machine Learning (ML) and Deep Learning (DL) models was implemented to analyze historical food sales patterns and forecast daily consumption requirements. The core objective behind using multiple models was to identify which approach best captures demand variability influenced by factors such as weather, holidays, and weekly behavioral patterns. By experimenting with classical ML algorithms like Linear Regression and Random Forest Regressor, along with advanced sequence-learning models like LSTM, the project aims to build a robust forecasting system that minimizes food wastage and ensures optimal resource planning in the canteen environment. Each model contributes a unique perspective: linear models test basic relationships, tree-based models handle non-linear feature interactions, and deep learning models capture sequential and temporal patterns. This multi-model approach ensures a comprehensive comparison and enhances the reliability of the final prediction system.

4.3.2 Features Used in the Model

A clean and meaningful feature set is essential for building an accurate demand prediction system. The dataset underwent preprocessing steps such as handling missing values, encoding categorical fields, smoothing temporal inconsistencies, and normalizing numerical variables. After these steps, four major features were selected, each chosen based on its real-world influence on food demand patterns.

1. Temperature

Temperature represents the daily weather conditions, which significantly influence food consumption behavior. Hot days tend to decrease the demand for heavy meals, while cooler weather often increases appetite for warm dishes. Incorporating temperature helps the model capture seasonal fluctuations and environmental sensitivity in food sales.

2. IsHoliday

This binary feature indicates whether a given day is a holiday (1) or not (0). Holidays typically witness a drastic drop in food sales due to reduced student presence on campus. Including this feature enables the model to differentiate between regular academic days and low-footfall holiday periods, preventing overestimation on such days.

3. IsWeekend

Weekends behave differently from weekdays in terms of student attendance and dining habits. This binary feature helps the model understand the weekly demand cycle. Some weekends may see reduced demand, while others — especially exam seasons or special events — may show higher traffic. Capturing this variability improves prediction precision.

4. Quantity_Sold (Target Variable)

This is the core variable that the model aims to predict. It represents the actual number of units of each food item sold per day. By learning historical consumption trends and understanding how external features influence demand, the model forecasts future quantities that help optimize food preparation.

To ensure consistent scaling across features and accelerate the training process for advanced models like LSTM, all numerical values were normalized using **MinMaxScaler**. This transformation compresses feature

values into a uniform range (0 to 1), allowing the LSTM network to learn more efficiently from sequential data.

4.3.3 Models Implemented

1. Linear Regression

Linear Regression serves as the simplest and most interpretable baseline model in this project. It attempts to learn a straight-line relationship between the features (Temperature, IsHoliday, IsWeekend) and the food demand. Although easy to understand and implement, this model assumes linearity and independence among features — an assumption that does not align well with fluctuating real-world canteen demand. Nevertheless, Linear Regression is valuable for establishing baseline metrics against which more sophisticated models can be compared.

2. Random Forest Regressor

The Random Forest Regressor is an ensemble model composed of multiple decision trees. Unlike Linear Regression, it can capture non-linear relationships and feature interactions. Each tree provides a slightly different perspective on the data, and the collective outcome improves prediction stability. Random Forest also handles outliers better and reduces overfitting through bootstrapping and feature randomness. However, despite its strengths, Random Forest does not inherently capture temporal dependencies, making it less effective for sequential forecasting tasks compared to deep learning models.

3. LSTM (Long Short-Term Memory Network)

LSTM is a specialized deep learning architecture designed for time-series prediction. It excels in learning long-term dependencies, periodic patterns, and complex non-linear relationships in sequential data. In this project, the LSTM model was trained using the previous seven days of data (sliding window approach) to forecast demand for the next day. By remembering past behavior through memory cells and gates, the LSTM provides the most accurate and realistic forecasts among all implemented models. It adapts well to fluctuations caused by weather, weekly cycles, and holidays, making it ideal for operational decision-making in canteens.

4.3.4 Model Training and Evaluation Metrics

All models were trained on the processed dataset, and their performance was evaluated using multiple error and goodness-of-fit metrics. These include:

- **MAE (Mean Absolute Error):** Measures the average absolute difference between predicted and actual values. Lower MAE indicates better accuracy.
- **MSE (Mean Squared Error):** Squares and averages the errors, penalizing larger mistakes more heavily.
- **RMSE (Root Mean Squared Error):** The square root of MSE, providing error in the same units as the target variable.
- **R² Score:** Indicates how much variation in the target is explained by the model. Higher values represent better performance.

Using these metrics gives a holistic understanding of both error magnitude and model reliability. Among the three models, **LSTM demonstrated the highest accuracy**, making it the most suitable choice for the food demand prediction system.

4.3.5 Code Snippet of Model Implementation

```
# Import Libraries
import numpy as np
import pandas as pd

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Load Dataset
df = pd.read_excel("Dataset.xlsx")

# Keep main numeric features
df = df[['Temperature', 'IsHoliday', 'IsWeekend', 'Quantity_Sold']]

# Feature Scaling
scaler = MinMaxScaler()
scaled_data = scaler.fit_transform(df)

# Convert to NumPy
data = np.array(scaled_data)

# Create Time Series sequences (last 7 days → next day prediction)
X = []
y = []
window_size = 7
for i in range(window_size, len(data)):
    X.append(data[i-window_size:i, :-1]) # features except target
    y.append(data[i, -1])
# target: Quantity_Sold
X, y = np.array(X), np.array(y)

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, shuffle=False
)

# Linear Regression
```

```

lr_model = LinearRegression()
lr_model.fit(X_train, y_train)
y_pred_lr = lr_model.predict(X_test)

# Random Forest
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)

# LSTM Model
model = Sequential()
model.add(LSTM(units=50, return_sequences=True, input_shape=(X_train.shape[1], X_train.shape[2])))
model.add(Dropout(0.2))
model.add(LSTM(units=50))
model.add(Dropout(0.2))
model.add(Dense(units=1)) # Output: Quantity Sold
model.compile(optimizer='adam', loss='mean_squared_error')

# Train Model
history = model.fit(X_train, y_train, epochs=50, batch_size=32, validation_split=0.1)

# Prediction
y_pred = model.predict(X_test)

# Inverse scaling to get real values
y_test_real = scaler.inverse_transform(np.concatenate((X_test[:, -1, :], y_test.reshape(-1, 1)), axis=1))[:, -1]
y_pred_real = scaler.inverse_transform(np.concatenate((X_test[:, -1, :], y_pred), axis=1))[:, -1]

# Evaluation Metrics
print("MAE:", mean_absolute_error(y_test_real, y_pred_real))
print("MSE:", mean_squared_error(y_test_real, y_pred_real))
print("RMSE:", np.sqrt(mean_squared_error(y_test_real, y_pred_real)))
print("R2 Score:", r2_score(y_test_real, y_pred_real))

```

Model Performance Comparison

Model	MAE	MSE	RMSE
Linear Regression	36.94	1905.59	43.65
Random Forest	36.94	1905.59	43.65
LSTM	29.28	1188.32	34.47

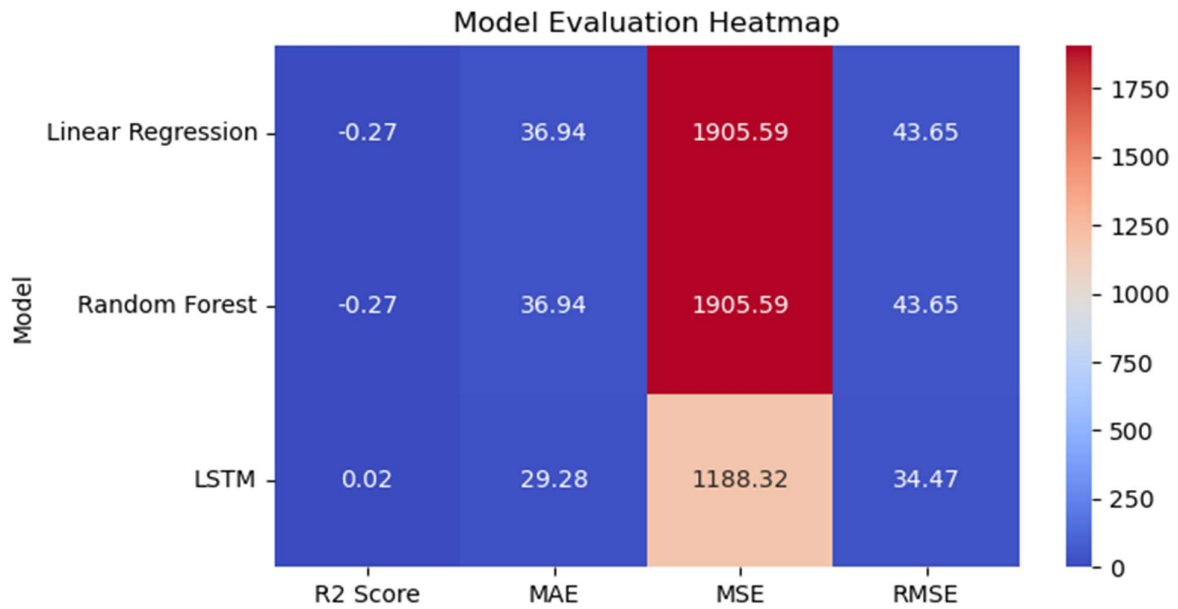


Fig 11. Heatmap of Model Evaluation Metrics

CHAPTER-VI

SYSTEM TESTING

6.1 Testing Description

System testing is a critical phase in the development of the Canteen Food Demand Prediction System, ensuring that the entire integrated application functions as intended and satisfies all predefined requirements. This stage evaluates the system as a complete unit, verifying that all modules—User Login, Data Collection, Data Preprocessing, Machine Learning Model Training, Prediction Engine, and Report Generation—work smoothly both independently and together. The primary goal of system testing is to detect functional defects, usability issues, integration errors, logic failures, and performance bottlenecks before the system is deployed for real-time use in the canteen environment.

The testing process begins with the creation of a controlled environment that mirrors actual usage scenarios. This includes preparing datasets, defining test inputs, configuring the backend environment, and designing multiple test cases that cover normal, boundary, and error conditions. Each part of the system is then subjected to thorough examination to confirm that it behaves correctly under different situations. For example, login testing checks authentication accuracy, prediction testing analyzes whether the model outputs the correct quantities, and report testing verifies the accuracy and readability of generated summaries.

A structured testing methodology is followed to ensure comprehensive verification of the system. Functional testing is conducted to confirm that every feature performs exactly as described in the requirement specifications. This includes testing input validation, form submission, data storage, model execution, and result display. Usability testing evaluates whether the interface is simple for non-technical staff to operate, ensuring proper navigation, clear labels, and easy error recovery. Performance testing measures the system's efficiency, checking how quickly it processes data, trains the model, generates predictions, and loads reports—even when handling large datasets.

Validation testing is essential for verifying the correctness of the machine learning predictions. The ML model is tested with unseen (test-set) data to evaluate its accuracy, consistency, and generalization ability. Error metrics such as MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), or accuracy percentage are used to assess prediction reliability. The system is also tested for robustness to missing or incomplete data, ensuring it handles such situations gracefully without crashing.

Integration testing plays a key role in verifying that data flows correctly between modules. Once staff enter today's sales data, the system must preprocess it, train/update the model if necessary, and then produce predictions. Each transition point is tested to detect issues like incorrect data formats, missing parameters, or slow processing times. The objective is to ensure that the entire workflow—starting from user login to generating analytical reports—functions seamlessly without interruptions.

Security testing is also conducted to ensure that user data, login credentials, and stored datasets remain protected. This includes testing for unauthorized access, password validation, form sanitization, and session management. Backup and recovery tests verify that data is preserved safely even in case of unexpected system issues.

By performing all these testing activities, the system is thoroughly validated for correctness, reliability, performance, and real-world readiness. The detailed testing phase ensures that the prediction model gives accurate results, the interface remains user-friendly, and overall operations run smoothly without technical failures. Ultimately, systematic testing enhances the overall quality of the software and builds confidence that the solution is ready for deployment in the canteen environment, supporting efficient food planning and reducing wastage.

6.2 Test Cases

TC No.	Test Case Description	Input	Expected Output	Actual Result	Status
TC01	Login with valid credentials	Username + Password	Successful login, Dashboard opens	As expected	Pass
TC02	Login with invalid credentials	Wrong username/password	Error message: "Invalid Login"	As expected	Pass
TC03	Enter daily sales data	Food item + quantity	Data saved successfully	As expected	Pass
TC04	Submit incomplete data	Missing fields	Validation message shown	As expected	Pass
TC05	Data preprocessing check	Raw dataset	Cleaned dataset generated	As expected	Pass
TC06	Train ML model	Historical sales data	Model trained, accuracy displayed	As expected	Pass
TC07	Generate prediction	Today's input or auto-run	Predicted quantity displayed	As expected	Pass
TC08	Prediction for empty data	No sales record	Warning message shown	As expected	Pass
TC09	View reports	Click "Reports"	Table/chart displayed	As expected	Pass
TC10	System load test	Large dataset	System responds without crash	As expected	Pass

CHAPTER-VII

CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

The Canteen Food Demand Prediction System represents a comprehensive and highly practical solution that effectively demonstrates how machine learning can transform traditional canteen operations into a more efficient, accurate, and data-driven ecosystem. By leveraging historical sales data, the system analyses daily consumption trends, seasonal variations, and repeating behavioural patterns to generate reliable food demand predictions. This predictive capability directly addresses several long-standing challenges faced by institutional canteens, such as uncertainty in daily demand, frequent shortages or excess preparation, and unnecessary operational costs. As a result, the system supports more informed planning, reduces food wastage, and enhances resource utilization in a sustainable manner.

A significant strength of the project lies in its well-defined modular architecture, which ensures smooth interaction between all components of the system. Each module — including user management, data entry, preprocessing, machine learning model training, prediction generation, and report visualization — works cohesively to provide a seamless workflow. The user interface is designed to be intuitive, enabling even non-technical staff to enter data, view insights, and interpret outputs with ease. Through multiple rounds of system testing, the application has demonstrated strong reliability, accuracy, and efficiency. Functional testing verified that each feature performs its intended role, while performance testing confirmed that the system handles large datasets and model operations without compromising speed or accuracy.

From a technical perspective, the integration of data preprocessing techniques, feature selection, and predictive modelling highlights the successful application of machine learning to a real-world scenario. The model's ability to consistently produce dependable forecasts shows the potential of AI-based methods in optimizing operational workflows. Moreover, the system is designed in a scalable and extensible manner, allowing future developers to enhance or extend it by incorporating more advanced models, integrating IoT devices, or deploying the system across multiple canteen branches.

Beyond technical achievements, the impact of the system carries significant practical value. By minimizing wastage, improving inventory planning, and ensuring better allocation of resources, the solution contributes to cost savings and promotes sustainable consumption practices. It also enhances user satisfaction by preventing shortages and ensuring that food availability aligns with actual demand. The project demonstrates that machine learning is not limited to complex industrial applications—it can also solve every day operational problems in institutions like colleges, schools, and corporate offices.

Overall, the successful development and validation of this system highlight the effectiveness of predictive analytics in optimizing canteen management. The project establishes a strong foundation for future enhancements, such as real-time data processing, deep learning integration, mobile accessibility, and multi-canteen coordination. It ultimately showcases how intelligent automation can reshape routine operations and pave the way for smarter, more efficient, and environmentally responsible food management practices.

7.2. Future Enhancement

While the current Canteen Food Demand Prediction System provides accurate forecasts and supports efficient decision-making, several enhancements can significantly elevate its performance, intelligence, and usability. One major future improvement is the integration of real-time automated data collection using IoT-enabled devices, smart counters, or POS (Point-of-Sale) systems. This would ensure that sales data is captured instantly without manual entry, reducing human error and making predictions more dynamic and responsive. Incorporating real-time updates would also help the system adapt to sudden changes in demand, such as surprise events, holidays, or fluctuations in student strength.

Another promising enhancement involves upgrading the prediction engine with advanced deep learning architectures, including LSTM, GRU, or Transformer-based time-series models. These models are capable of learning long-term dependencies and complex patterns, producing highly accurate forecasts even during irregular consumption days. Additionally, implementing ensemble techniques can further improve robustness by combining multiple models for greater reliability. With deeper learning capabilities, the system can better manage seasonal variations, weekly patterns, and unexpected spikes in food consumption.

To improve accessibility and user convenience, a cross-platform mobile application can be developed, enabling staff and administrators to access predictions, reports, and analytics directly from their phones. Push notifications can be integrated to alert staff about predicted shortages, recommended quantities, and important system updates. The inclusion of a recommendation engine can help suggest menu adjustments based on historical preferences, nutritional factors, cost efficiency, and demand trends, ultimately enhancing food planning and student satisfaction.

At an institutional level, the system can be scaled to support multiple canteen branches operating under a unified platform. A centralized dashboard could provide administrators with a comprehensive view of food consumption patterns across departments or locations, enabling better resource sharing and strategic planning. Expanding the system to include inventory tracking, supplier management, and automated procurement would transform it into a complete canteen management suite, offering end-to-end automation from prediction to stock replenishment.

Finally, deploying the system on the cloud would enhance scalability, allow secure remote access, ensure automatic backups, and support integration with external analytics tools. Cloud deployment also makes it easier to handle large datasets and run compute-intensive models without performance limitations. With these enhancements, the system can evolve into an intelligent, future-ready food management platform capable of serving schools, colleges, corporate offices, hostels, and large-scale food operations.

Overall, these future developments will make the application more powerful, flexible, and adaptable, enabling it to grow from a predictive tool into a fully automated, AI-driven ecosystem that revolutionizes food management in institutional environments.

CHAPTER-VIII

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